Computational Physics Lectures

Week 03: Optimization

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Week 3

Optimization

3.1 Introduction

In many applications of computational physics, one is interested not only in thermal averages but also in identifying particularly stable configurations of a system. Examples include determining the equilibrium geometry of a molecule, the crystal structure of a material, or the lowest-energy arrangement of atoms on a surface. Such problems naturally lead to optimization.

As discussed in the previous chapter, the Boltzmann factor, $\exp(-V(x)/(k_B T))$, strongly favors regions of low potential energy. The thermal probability distribution may therefore be highly peaked around the minima of the potential-energy landscape. Depending on the physical system and the properties of interest, it may be useful to identify these minima before studying the system further.

The minimum with the lowest energy is called the *global minimum*, while the other minima are referred to as *local minima*. This chapter concerns methods for determining both local and global minima. These are naturally called methods for local and global optimization. Other commonly used terms for local optimization are *relaxation* and *quenching*.

There are very many optimization methods, and those selected here represent only a small subset of the available methods.

3.2 Metropolis Monte Carlo search

The acceptance criterion in the Metropolis Monte Carlo method reads

$$\mathrm{RND} < \min\left(1, \exp(-\Delta E/(k_B T))\right),$$

i.e. a uniformly distributed random number between 0 and 1 must be smaller than $\min\left(1, \exp(-\Delta E/(k_B T))\right)$. The criterion can equivalently be written as

$$\mathrm{RND} < \exp(-\Delta E/(k_B T)), \tag{3.1}$$

since RND can never be larger than 1.

When written in the latter form, Eq. (3.1), the criterion is easy to interpret. Steps that decrease the energy are always accepted, while steps that increase the energy are accepted with a probability that

decreases exponentially with the energy increase. Depending on the temperature, a Metropolis Monte Carlo trajectory can therefore move through an energy landscape and overcome energy barriers separating regions containing different local minima. This may form the basis of a global optimization method, the so-called Monte Carlo search method. One or several trajectories are generated with the available computational resources, and the lowest-energy structure observed during the search is reported as the *putative global minimum*, since one generally cannot prove that no lower-energy structure exists.

In the limit $T \to 0$, all moves that increase the energy are rejected. The trajectory therefore performs a downhill search and eventually becomes trapped in a local minimum. In this limit, the method may be regarded as a local optimization method. At zero temperature, it can be characterized as random optimization, with the potential energy as the objective function and the positional coordinates defining the optimization space.

For a high-dimensional problem, the Metropolis Monte Carlo search may benefit from trial configurations in which only a subset of the coordinates is changed at a time rather than all coordinates simultaneously.

3.3 Basins of attraction

Suppose that a local optimization method is started from different initial configurations. Many distinct starting configurations may converge to the same local minimum. The set of initial configurations that converge to a given local minimum under a particular local optimization method is called the *basin of attraction* of that minimum.

The basins depend not only on the energy landscape but also on the local optimization method. Global optimization may be viewed as the task of exploring different basins of attraction and identifying the basin containing the global minimum.

3.4 Random Structure Search

If a method for local optimization is available, a very simple method for global optimization can be formulated, namely the *Random Structure Search* method. It consists of the following steps:

1. Generate the system in a random configuration. For a one-dimensional system, this amounts to selecting a random coordinate x. For a d-dimensional system, it corresponds to selecting a random vector $\mathbf{r} \in \mathbb{R}^d$.
2. Perform a local optimization of the system.
3. Record the optimized configuration if it has the lowest energy encountered so far.
4. Repeat steps 1–3 as many times as computationally affordable.
5. Report the lowest-energy configuration found.

Each random starting configuration has some probability of lying in the basin of attraction of the global minimum. Repeating the procedure increases the probability of finding that basin.

However, a minimum with a very narrow basin of attraction may be difficult to locate by random sampling.

3.5 Gradient-based local optimization

The zero-temperature Metropolis Monte Carlo search introduced above (Section 3.2) may be useful as a local optimization method when the objective function, in this context the potential energy, is the only available information. Often, however, the gradient or higher-order derivatives of the objective function are available at a low computational cost. More efficient optimization methods may then be formulated.

When the objective function is the potential energy, the force $\mathbf{F}$ on the constituents of the system is the negative gradient of the energy:

$$\mathbf{F}(\mathbf{r}) = -\nabla E(\mathbf{r}),$$

where E is the total potential energy and $\mathbf{r}$ is a vector containing all positional coordinates. The gradient points in the direction in which the energy increases most rapidly. The force therefore points in the direction of the most rapid local decrease in energy.

3.6 Steepest descent and gradient descent

Whenever the force is available, a local optimization method can be formulated by taking small steps in the direction of the force. In these notes, we use the term *steepest descent* for the following fixed-step procedure:

1. Start at $\mathbf{r}$.
2. Calculate the direction of the force:
$$\hat{\mathbf{F}}(\mathbf{r}) = \frac{\mathbf{F}(\mathbf{r})}{\|\mathbf{F}(\mathbf{r})\|}.$$
3. Take a fixed-length step in the direction of the force:
$$\mathbf{r} \to \mathbf{r} + \Delta\hat{\mathbf{F}}(\mathbf{r}),$$
where Δ is a predetermined step length.
4. Repeat steps 2–3 until a chosen convergence criterion is satisfied.

In the limit $\Delta \to 0$, the method formally traces a path in the steepest-descent direction, which motivates the name. An obvious difficulty is that a very small but finite value of Δ requires many steps. Moreover, the optimization continues to take steps of length Δ even as the force approaches zero.

A slightly modified procedure is the *gradient descent* method, in which the update is

$$\mathbf{r} \to \mathbf{r} + \gamma\mathbf{F}(\mathbf{r}),$$

where γ is a constant. The step length now decreases together with the force as the optimization converges. However, excessively large steps may be taken in regions where the force is large. This can be remedied by imposing a maximum step length.

The terminology varies in the literature, and both procedures are frequently referred to as gradient descent. The essential distinction here is that the first procedure normalizes the force and uses a fixed step length, whereas the second uses a step proportional to the force magnitude.

3.7 Line search

The difficulties associated with choosing suitable step lengths in steepest descent and gradient descent can be reduced by making fuller use of a search direction once it has been established. A line search reduces the multidimensional optimization problem to a one-dimensional problem along the chosen direction. It consists of the following steps:

1. Start at $\mathbf{r}$.
2. Calculate the direction of the force:
$$\hat{\mathbf{F}}(\mathbf{r}) = \frac{\mathbf{F}(\mathbf{r})}{\|\mathbf{F}(\mathbf{r})\|}.$$
3. Define the energy along that direction:
$$\widetilde{E}(\alpha) = E\left(\mathbf{r} + \alpha\hat{\mathbf{F}}(\mathbf{r})\right).$$
4. Find α_{opt} that minimizes $\widetilde{E}(\alpha)$ and update the configuration according to
$$\mathbf{r} \rightarrow \mathbf{r} + \alpha_{\text{opt}}\hat{\mathbf{F}}(\mathbf{r}).$$
5. Repeat steps 2–4 until the magnitude of the force is below a prescribed tolerance or another convergence criterion is satisfied.

An exact minimization along the line is not always necessary. In practice, it is often sufficient to find a step that gives an adequate decrease in the energy while avoiding an excessive number of energy and force evaluations.

3.8 Second-order methods

The gradient provides information about the local slope of the energy landscape. Additional information about its local curvature is contained in the Hessian matrix,

$$H_{ij}(\mathbf{r}) = \frac{\partial^2 E(\mathbf{r})}{\partial r_i \partial r_j}.$$

In two dimensions, the Hessian has the form

$$\mathbf{H}(\mathbf{r}) = \nabla^2 E(\mathbf{r}) = \begin{pmatrix} \partial^2 E/\partial x^2 & \partial^2 E/\partial x \partial y \\ \partial^2 E/\partial y \partial x & \partial^2 E/\partial y^2 \end{pmatrix}.$$

Near a minimum, the energy can be approximated by a quadratic expansion:

$$E(\mathbf{r}_k + \boldsymbol{\Delta}) \approx E(\mathbf{r}_k) + \nabla E(\mathbf{r}_k)^\mathsf{T} \boldsymbol{\Delta} + \frac{1}{2} \boldsymbol{\Delta}^\mathsf{T} \mathbf{H}(\mathbf{r}_k) \boldsymbol{\Delta}.$$

The minimum of this local quadratic model is found by differentiating with respect to $\boldsymbol{\Delta}$ and setting the result equal to zero:

$$\nabla E(\mathbf{r}_k) + \mathbf{H}(\mathbf{r}_k) \boldsymbol{\Delta}_k = 0.$$

Thus, the Newton step is obtained by solving

$$\boxed{\mathbf{H}(\mathbf{r}_k) \boldsymbol{\Delta}_k = -\nabla E(\mathbf{r}_k)}, \qquad \boxed{\mathbf{r}_{k+1} = \mathbf{r}_k + \boldsymbol{\Delta}_k}.$$

In numerical work, one should solve this linear system rather than explicitly calculate $\mathbf{H}^{-1}$. In Python it would look something like:

```
def newton_method(r0, tolerance=1e-10, max_iterations=20):
    """Return all iterates produced by undamped Newton steps."""
    points = [np.asarray(r0, dtype=float)]
    for _ in range(max_iterations):
        rk = points[-1]
        # Solve H Delta = -g; do not form the inverse of H explicitly.
        delta = np.linalg.solve(hessian(rk), -gradient(rk))
        points.append(rk + delta)
        if np.linalg.norm(delta) < tolerance:
            break
    return np.vstack(points)
```

where `hessian` and `gradient` are functions defined elsewhere.

In strict mathematical terms, we can say that if the Hessian is positive definite, the local quadratic approximation has a unique minimum, and minimizing it gives the Newton step

$$\mathbf{r} \to \mathbf{r} - \mathbf{H}^{-1}(\mathbf{r}) \nabla E(\mathbf{r}) = \mathbf{r} + \mathbf{H}^{-1}(\mathbf{r}) \mathbf{F}(\mathbf{r}).$$

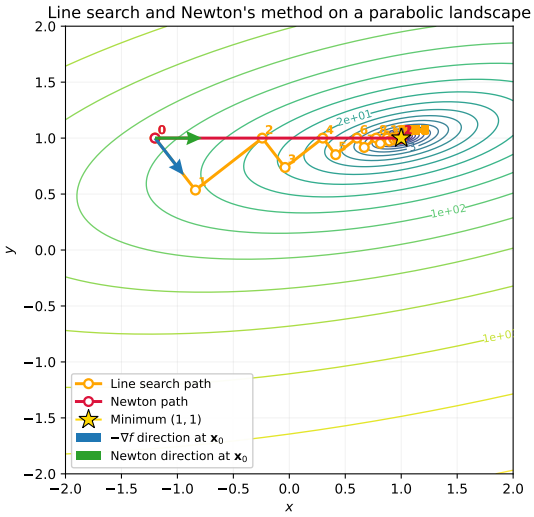
Notice that both the direction and length of the Newton step are determined entirely by the gradient and Hessian, rather than involving a user-defined step-size parameter.

Unlike gradient descent or gradient-based line search, the Newton step accounts for differences in curvature between directions. Close to a minimum, Newton's method can therefore converge much faster than a first-order method.

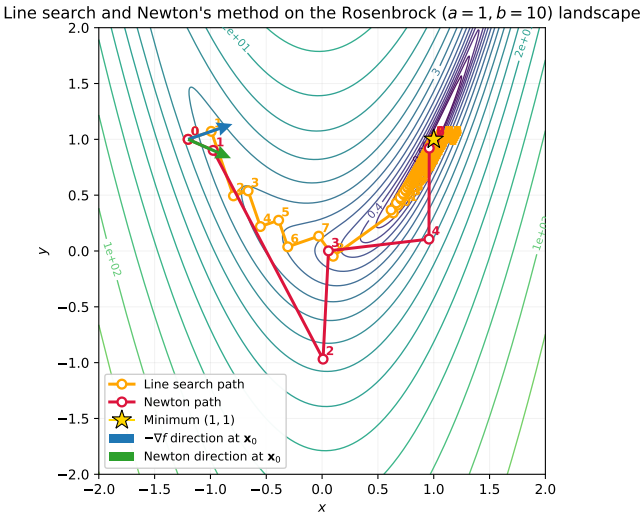
Calculating and storing the complete Hessian may, however, be expensive for a system with many coordinates. In addition, far from a minimum the Hessian need not be positive definite, and the Newton direction is then not necessarily a downhill direction. These issues motivate *quasi-Newton methods*, which construct an approximation to the Hessian or its inverse from successive positions and gradients. The BFGS method is a widely used example. In practice, Newton and quasi-Newton directions are often combined with a line search or a maximum allowed step length.

Examples

If the energy landscape is two-dimensional and harmonic, the Newton method finds the minimum in exactly one step, while a line search may need very many steps as shown in this example:



For more complicated energy landscapes with narrow valleys the Newton method requires more iterations, but still converges much faster than a gradient-based line search. Here is an example:



3.9 Basin Hopping

The basin hopping method is the Metropolis Monte Carlo search method augmented with a local optimization at every step. It takes the following form:

1. Start at $\mathbf{r} \in \mathbb{R}^d$ and locally optimize the initial configuration.
2. Set up a trial configuration by making a random step:

   $$\mathbf{r}' = \mathbf{r} + \mathbf{r}_{\mathrm{RND}},$$

   where $\mathbf{r}_{\mathrm{RND}} \in \mathbb{R}^d$ has normally distributed values for each coordinate.
3. Relax the trial configuration:

   $$\mathbf{r}' \to \widetilde{\mathbf{r}}',$$

   where $\mathbf{F}(\widetilde{\mathbf{r}}') = 0$ to within a prescribed convergence tolerance.
4. Accept the move $\mathbf{r} \to \widetilde{\mathbf{r}}'$ if the Metropolis Monte Carlo criterion is fulfilled:

   $$\mathrm{RND} < \exp\left(-\frac{E(\widetilde{\mathbf{r}}') - E(\mathbf{r})}{k_B T}\right),$$

   where RND is a random number uniformly distributed between 0 and 1. If the move is rejected, retain $\mathbf{r}$ as the current configuration.
5. Repeat steps 2–4 as many times as computationally affordable.

Because every trial configuration is locally optimized before the acceptance criterion is applied, basin hopping performs a Monte Carlo walk between local minima rather than between arbitrary configurations. The original energy landscape is effectively transformed into a landscape in which every point in a basin of attraction is assigned the energy of the corresponding local minimum. The algorithm then hops between these basins, which explains the name *basin hopping*.

The temperature in basin hopping is an algorithmic parameter controlling the probability of accepting moves to higher-energy minima. It need not represent the physical temperature of the system. Basin hopping combines local optimization with stochastic exploration and is therefore useful for many global optimization problems in computational physics and chemistry.